Problem Statement

App Developer

" Developer: I don't have a easy way to search for problems in my application infrastructure"
" OPS: I don't want to look at a thousand screens to understand if my stack is healthy"
" E-Commerce: I want to know the cost of slow performance on my end users"
" Management/Finance: We plan capacity in a ad-hoc fashion, developers double machines on every marketing event to be on the safe side, I need a better way"

Infrared >>

Intelligent Monitoring and Operational Analytics for Mobile Apps that is built around Application Topology Awareness

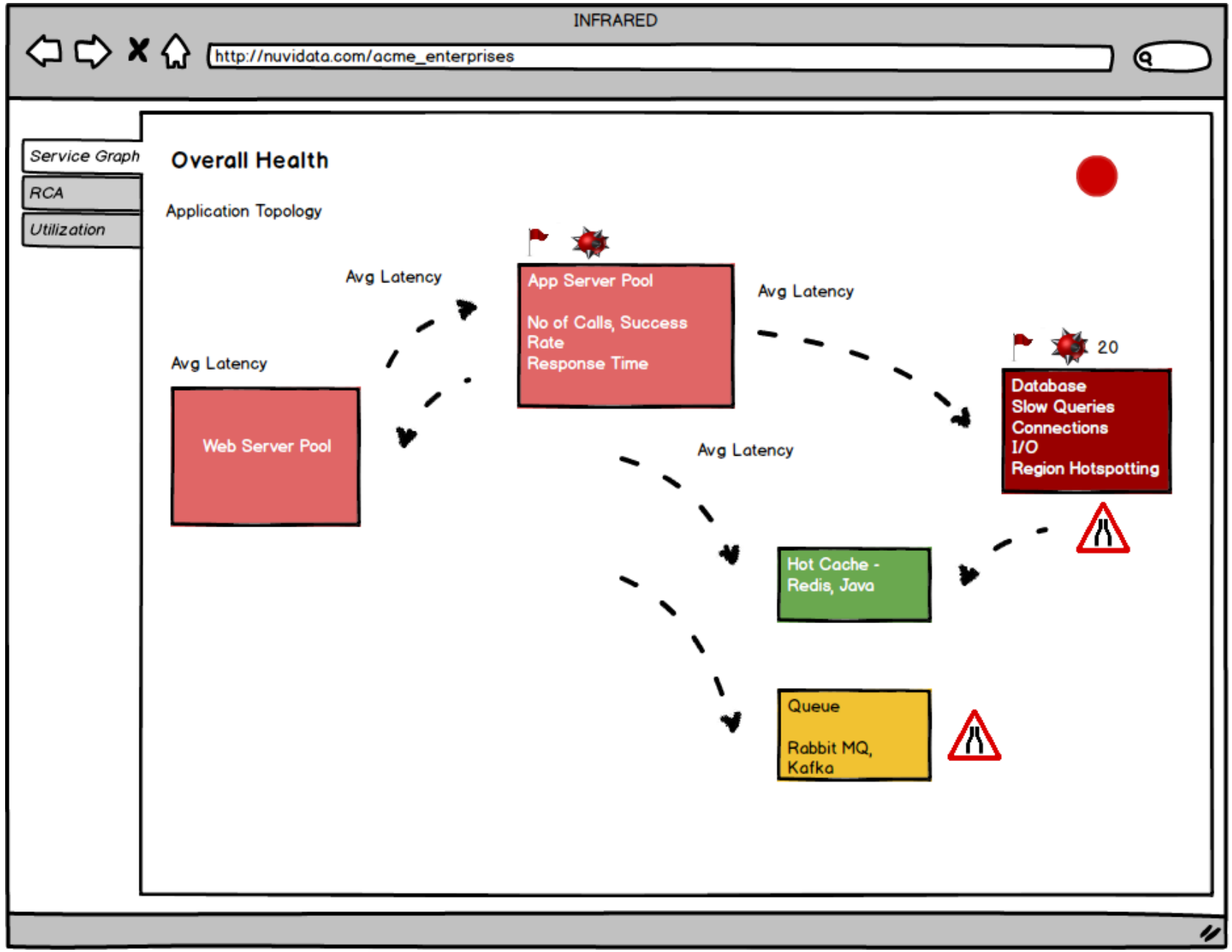
A system that combines metrics, logs with big data and machine learning to help you debug, check performance and plan capacity. Uses ML algorithms and classifiers to build recommendations, points to anomalies, suppress noise and reveal important attributes about your app in realtime

Features:
End to end view of the application stack
Completely scalable, Can support thousands of nodes
Quick Deployment over SAAS, No code level instrumentation required, AWS and AZURE support

Monitor the health of a service
- Host level system metrics
- process level systemic metrics

Simple Root Cause Analysis. No need to look at a per node montoring screens and alerts

Phase 2 : Application Performance
Phase 3 : Capacity Planning & Predictive Analytics
- Performance bottlenecks based on Seasonal Variation
- Errors by Severity



Icon List

Bottleneck

Critical

Warning

SPF

INFRARED

http://nuvidata.com/acme_enterprises

- Service Categories
  - Web Server
  - App Server (10 I) 2C 3W
    - Instance 1 (1C)
    - Instance 2
    - Instance 3 (2C)
  - Queue
  - Cache
  - RDBMS
  - NoSQL

Events per hour

2W 1M

1h
2h
23h

Mon Tue Wed Thur Fri

Severity	Class	Type	Detail	Instances	Date	Mark Resolved
Critical	Anomaly	Spike	Extended CPU Usage detected ongoing past 10 mins Add Alert	Instance-1 Instance-3	12th April 3:am	☐
Critical	Anomaly	Resource Exhaustion	Memory usage is growing @ 2M per min for last 10 mins	Instance-2	13th April 1pm	☐

Important Metrics (Cumulative)

Anomaly

Deploy Event : User: Rick

INFRARED

http://nuvidata.com/acme_enterprises

Severity	Class	Type	Detail	Instances	Date	Mark Resolved
Critical	Anomaly	Spike	Extended CPU Usage detected ongoing past 10 mins	Instance-1 Instance-3	12th April 3:am	☐
Critical	Anomaly	Resource Exhaustion	Memory usage is growing @ 2M per min for last 10 mins	Instance-2	13th April 1pm	☐

Important Metrics (Instance 1)

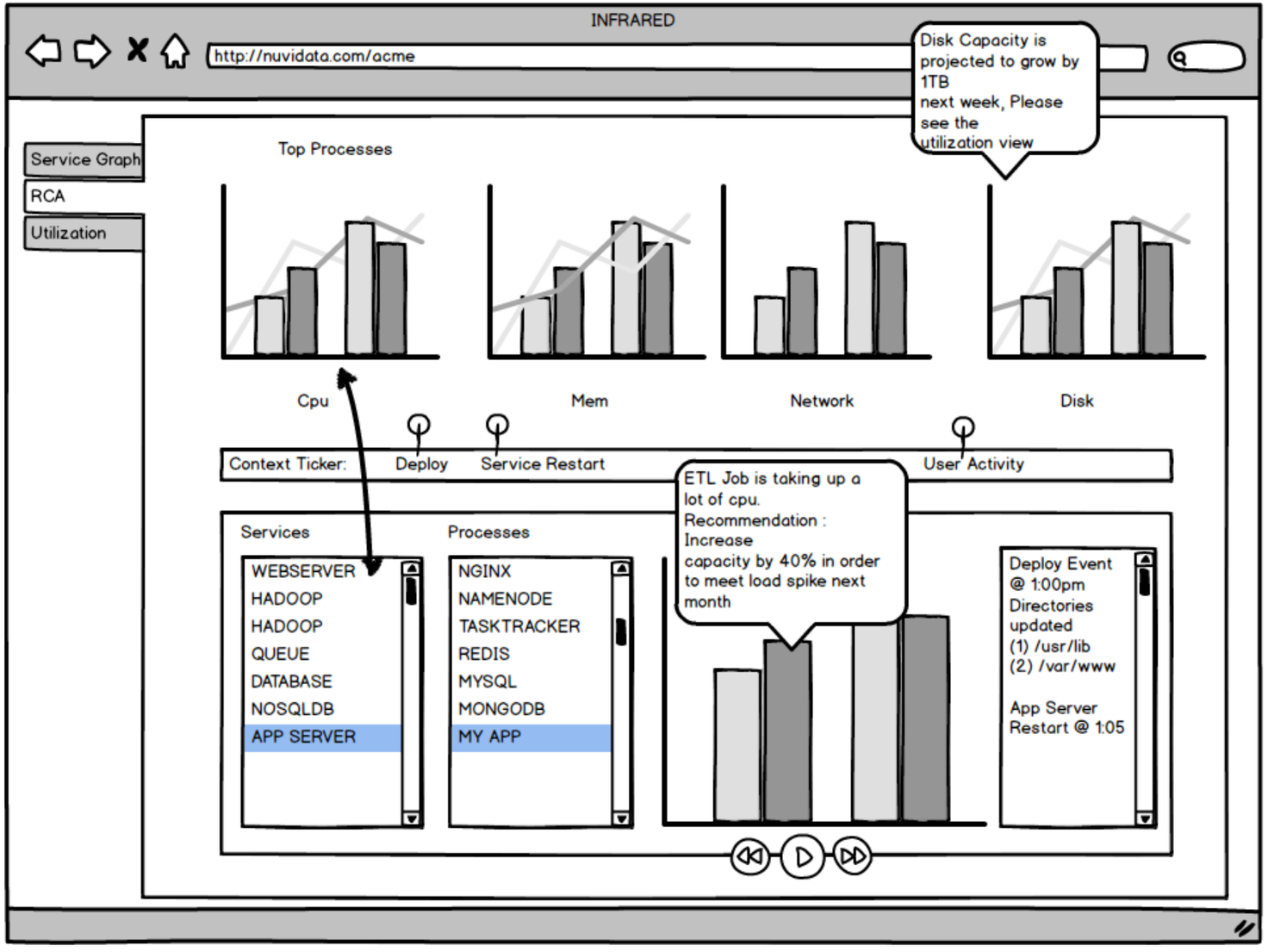
Deploy Event : User: Rick

System

Anomaly

Service Specific

Custom



Awareness Bubbles

By Service

Memory Semantics
- GC

File growth rate Vs Capacity
- Known logging locations
- Sensistive paths like root volumes

Service and Process Restarts

Change Events
For example: Install events, Deployments
Configuration / File system Changes

CPU Usage
- Hadoop : Jobs , I/O patterns, Hotspots, Compaction
- Redis : Key R/W volume
- App Server : Business Functions by time taken
- Database : Top slow queries

Configuration Optimizer

Big Data
Hadoop
HDInsight

Web Tier
NgInx
IIS

Queues
RabbitMQ
Kafka

Caches
Redis
Memcached

Databases
MySQL
Oracle
MongoDB
Couchbase
Hbase
Cassandra

INFRARED

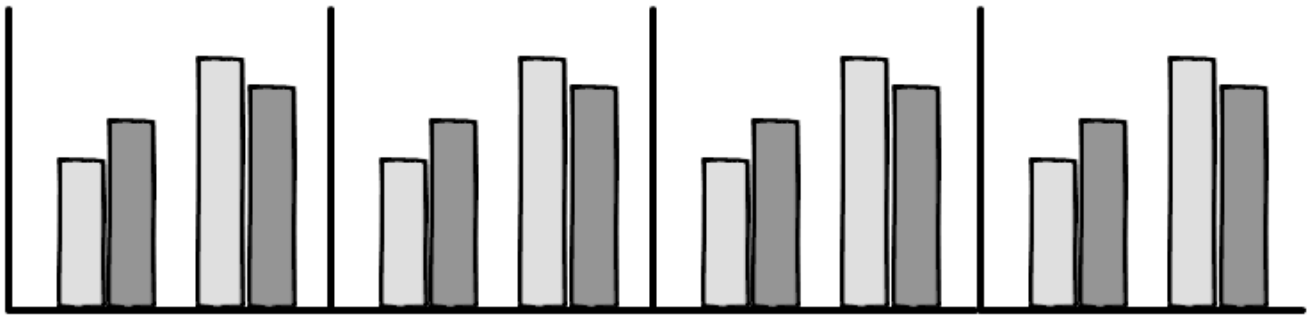
http://nuvidata.com/acme

Service Graph

RCA

Utilization

Search : NullPointerException host=track-02

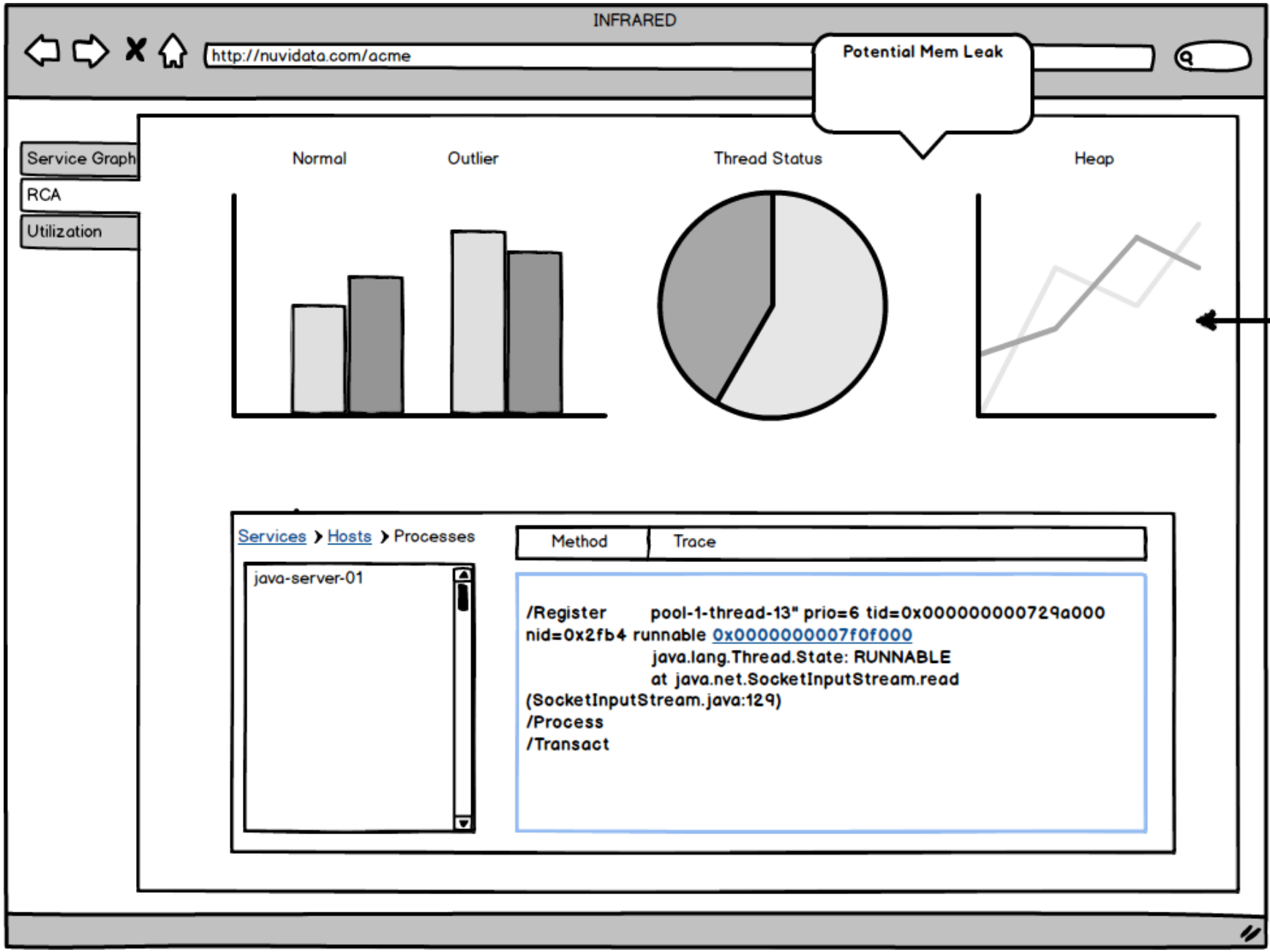


Services › Processes › Hosts

- App-Track-01
- App-Track-02
- App-Track-03
- App-Track-04
- App-Track-05
- App-Track-06
- App-Track-07
- App-Track-08

Logs

```
java.lang.Throwable
        at com.intellij.openapi.util.TraceableDisposable.kill
(TraceableDisposable.java:41)
        at com.intellij.openapi.editor.impl.EditorImpl.release
(EditorImpl.java:815)
ERROR - api.util.objectTree.ObjectNode - JDK: 1.7.0_80
NullPointerException
```



Sawtooth Pattern - Ideal GC